

README

QA evidence — qBittorrent WebUI login repair + themed-overlay removal

Revision: 2 **Last modified:** 2026-09-02T05:50:00Z

Captured evidence for the 2026-09-01 repair of “cannot log in to the qBittorrent WebUI with admin/admin”, plus the boot failure, the themed-WebUI overlay removal, and the systemd reboot-survival work.

Revision 2 (2026-09-02) adds the round-2 exhaustive root-cause sweep — see the final section. The original round-1 text above is preserved unedited; every status change is recorded as a change rather than rewritten in place, so the document shows what was believed when (§11.4.6).

Image under test: `lscr.io/linuxserver/qbittorrent:latest` → **qBittorrent v5.2.3 / WebAPI 2.15.1** (measured, not assumed).

The defect chain (all proven, none inferred)

#	Defect	How it was proven
RC-1	Credentials authored into <code>config/qBittorrent/config/qBittorrent.conf</code> , but the image reads <code>/config/qBittorrent/qBittorrent.conf</code> and launches <code>qbittorrent-nox</code> with no --profile	Read the image’s own <code>init-qbittorrent-config/run</code> and <code>svc-qbittorrent/run</code>
RC-2		

#	Defect	How it was proven
	WEBUI_USERNAME / WEBUI_PASSWORD in compose are inert — the image implements neither	Control needle: same grep over the image found WEBUI_PORT, found nothing for those two
RC-3	qBittorrent 5.x therefore mints a random temporary password every boot	runs/ RED_temporary_password_capture.log
	start.sh	
RC-4	_ensure_webui_credentials was replace-only (grep -q ... && sed), so it wrote nothing into a config lacking the keys	runs/start_sh_boot_integrity.txt (D2), RED then GREEN
RC-5	Every password path detected success by body == "0k."; 5.2.3 returns 204 + empty body + QBT_SID_<port>	Measured live; runs/ live_verification.txt
RC-6	Go client required 200 and "0k." and cookie SID — could never authenticate	qBitTorrent-go/internal/client/ auth_qbt5_test.go, RED then GREEN
RC-7	Boot abort: ((attempt++)) with attempt=0 returns exit 1 under set -euo pipefail, killing start.sh at the first retry	runs/start_sh_boot_integrity.txt (D1)
RC-8	cleanup_stale_config deleted the live config every boot — its predicate matched qBittorrent's own normalised SavePath=/downloads/	3 accumulated .backup.* files; predicate matched twice
RC-9	Security regression caught during the repair: enforcing LocalHostAuth=false + AuthSubnetWhitelistEnabled=true disables auth for loopback and all RFC1918. A wrong password returned 204 (accepted)	Negative control in runs/ live_verification.txt
RC-10	Angular dashboard never built → _angular_available=False → :7187/ served {"dashboard":"not found"} under HTTP 200	runs/final_verification.txt before/after

Files

File	What it holds
runs/ RED_temporary_password_capture.log	The original qBittorrent boot log showing the temporary password being minted
runs/start_sh_boot_integrity.txt	GREEN run of the D1–D4 boot-integrity guard
runs/cm_gate.txt	CM-QBITTORRENT-WEBUI - CREDENTIALS gate result
runs/cm_gate_mutation.txt	Paired §1.1 mutation meta-test — gate fires on 4 broken arms, clean on 3 good arms
runs/full_boot.log	Complete ./start.sh boot, ANSI-stripped, with every warning
runs/live_verification.txt	Login matrix + vanilla-WebUI integrity immediately after the fix
runs/final_verification.txt	Full-stack sweep: containers, ports, login, dashboard, security posture, systemd

Honest boundary (§11.4.6)

PROVEN by captured runtime evidence: - admin/admin authenticates on both :7185 and :7186 (HTTP 204 + QBT_SID) - A wrong password is **rejected** (401) and an unauthenticated API call is Forbidden — authentication is real, not bypassed - No temporary password is minted - All four containers healthy; vanilla qBittorrent WebUI served with zero overlay - Angular dashboard serves with assets resolving - systemd units installed, boba.target enabled, Linger=yes

NOT PROVEN — stated as owed, not claimed: - Reboot survival end-to-end. Units are installed, enabled, valid, and linger is on — necessary, *not sufficient*. A real reboot is the only proof and host power operations are hard-banned here (CONST-033). Operator-attended verification required. - **Firefox / WebKit**. The browser proof ran on Chromium only. - **docs/qa/ historical twins**. 698 .html/.pdf files were re-rendered

by a pandoc 3.10 → 3.11 upgrade even though their .md sources never changed; 320 of those are historical evidence artifacts. Included at operator instruction after the alternative was raised.

RESOLVED SINCE THIS FILE WAS FIRST WRITTEN (recorded rather than silently edited, so the record shows what was true when): - ~~pytest not installed~~ — uv was available all along; a venv exists at .venv and the full unit sweep runs. My earlier “needs python3-venv + sudo” conclusion was wrong: I had checked one mechanism, not the alternatives. - ~~Playwright not installed~~ — it arrived with the frontend npm install. The vanilla-WebUI claim now rests on a REAL Chromium render: window.qBittorrent live with 27 members and 17 Client functions, zero console errors, and live sync data (Free space: 1.533 TiB, DHT: 122 nodes) visible in the captured screenshot — not on HTML-integrity assertions. - ~~doc-twin-exporter not runnable~~ — scripts/generate_markdown_exports.sh existed and was already wired into pre-build; it was gated on a from weasyprint import HTML import while execing the weasyprint CLI, so it printed “PDF support: not available” and exited 0 having written nothing.

Owed tracked items (§11.4.197) — ALL THREE NOW RESOLVED (2026-09-02)

Status change. These three were staged here on 2026-09-01 because the mandated minting path was unavailable (no built workable-items binary, no consumer config for constitution/scripts/reporting/report_item.sh), and hand-writing rows into docs/Issues.md would desynchronise the SQLite single-source-of-truth (§11.4.93/§11.4.95).

On 2026-09-02 the operator directed that **everything discovered be systematically investigated and fixed at root cause — nothing parked as a ticket**. All three were therefore closed rather than tracked. Their original text is preserved below unedited, each followed by its resolution, so the record shows what was true when (§11.4.6 — resolution recorded, not silently rewritten).

The workable-items binary was subsequently built during this work (it is gitignored), which is also why the docs-sync commit seam’s CHECK 1-3 had not been running at all.

Resolution summary:

Item	Resolution
OWED-1 shared add-success predicate	CLOSED — download-proxy/src/merge_service/qbit_add.py; one mutation now fails BOTH consumers (5 tests across the bridge and routes halves), verified and sha256-restored
OWED-2 stress-suite torrent purge	CLOSED — diff-scoped to after - before; loud on failure; blocking gate 55 CM-NO-UNSCOPED-LIVE-DESTRUCTION + 11-arm meta-test + 5 hermetic regression tests
OWED-3 two standing test reds	CLOSED — test_no_runtime_service_skips (3 fixture gates + 2 # allow-skip: annotations, the latter carrier-matched merge_service/ as a service); test_bob129_slowapi_response_contract (import-alias duplicate registration under api_routes, fixed in two independent layers)

OWED-1 — Extract a shared `_qbit_add_succeeded` predicate (Type: Task)

Two implementations of qBittorrent’s add-success contract now exist: download-proxy/src/api/routes.py: `_qbit_add_succeeded` and webui-bridge.py: `_qbit_add_succeeded`. Review #3 findings F2/F3/F4 all trace to that duplication — the two copies drifted and briefly recorded *opposite* 409 verdicts. Extract one dependency-free module both import (§11.4.251).

Acceptance: a single mutation of the shared logic must make BOTH sites’ tests fail. Today a mutation of either copy leaves the other’s tests green, which is exactly how the divergence went unnoticed.

Measured contract the shared predicate must encode (qBittorrent v5.2.3 / WebAPI 2.15.1, 2026-09-01):

case	response	meaning
valid file add	200 { "success_count": 1 }	added
.torrent URL add		added, fetching async

case	response	meaning
	202 {"pending_count":1}	
magnet add	200 {"success_count":1}	added
duplicate add	409 Conflict	already present (success)
no payload	409 Conflict	nothing added (failure)
corrupt/truncated/ empty file	415	rejected

The 409 ambiguity is the load-bearing part: it means success OR failure depending on whether a payload was sent, and the status alone cannot tell. Empty URLs are now rejected at `DownloadRequest` validation so a 409 reaching the predicate implies a genuine duplicate — the shared version must preserve that invariant or restore the ambiguity explicitly.

OWED-2 — Scope the stress suite’s torrent purge (Type: Bug)

tests/stress/test_search_stress.py:18-54
`_purge_qbittorrent_torrents()` lists **every torrent in the instance** via `torrents/info` — not diff-scoped — and deletes them all with `deleteFiles=false`, from an autouse fixture that runs before every stress test. On an operator’s live instance that silently de-registers their entire library (files survive; the session does not).

Pre-existing (landed a684b2f, 2026-04-20), so not caused by this batch, and it did NOT cause the 2026-09-01 incident — that one’s log signature is staggered single removals with a WebUI page-load six seconds prior, whereas this purge issues one batched delete. But the hazard is real and standing.

Fix: scope to test-added hashes only, as the bridge suite already does (added = `hashes_after - before`). Never “delete everything the operator owns”.

Also a §11.4.238 coverage escape: an agent found this, no gate did.

OWED-3 — Two standing test reds (Type: Task)

- tests/unit/test_no_runtime_service_skips.py — 5 service-availability skips in tests/scaling/ and tests/unit/test_plugin_rutracker.py need converting to fixture gates. Pre-existing; fails at HEAD too.
 - tests/unit/test_bob129_slowapi_response_contract.py — fails in the full sweep, PASSES in isolation. Order-dependent or flaky; cause UNKNOWN, needs §11.4.102 systematic debugging. §11.4.248 quarantine-or-stabilise candidate.
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Round 2 (2026-09-02) — exhaustive root-cause sweep

Operator instruction: *“everything discovered MUST BE exhaustively and systematically investigated, debugged and fixed by addressing the real root causes of all issues!”*

Every item below was found DURING this work, not reported by the operator — which per §11.4.238 makes each one a coverage escape in its own right. Each was root-caused, fixed, and paired with a mutation proving the fix load-bearing.

Security / data-safety

Defect	Root cause	Proof
ci.sh secret scan structurally blind	PCRE (?:...) under grep -E (ERE), warning eaten by 2>/dev/null; ALSO ["\x27] in a bracket expression is five literal chars, so '-quoted secrets could never match	planted needle FAILs the step; both quote styles now detected; control needle assembled at runtime so ci.sh cannot match itself
.env truncation → master-key loss	grep ... \\ \\ true then unconditional mv -f; empty temp published over .env	line-count + key-NAME-set invariants; aborts loudly, .env byte-identical; proven against the silent-drop case an

Defect	Root cause	Proof
Unscoped live destruction	<code>"\ ".join(t["hash"] for t in torrents)</code> from an autouse fixture — every operator torrent, <code>deleteFiles=false</code>	exit-code check misses diff-scoped to after - before; gate 55 + 11-arm meta- test; None baseline deliberately disarms teardown rather than degrading to <code>set()</code> (which would delete everything)
Jackett API key in ps-visible argv	key interpolated into a <code>python -c</code> command line	moved to <code>BOBA_JACKETT_KEY</code> env; key value appears 0 times in all output
<code>start.sh</code> two silent write paths	<code>\\ \\ </code> true on the plugin JSON, AND <code>sed_inplace</code> ending in <code>rm -f ...bobabak</code> which succeeds after a failed <code>sed -i</code>	attempt → verify by read-back → report; [SUCCESS] no longer printable on a write that never landed

Correctness

Defect	Root cause
7 rutracker tests order- dependent	<code>_load_rutracker()</code> set <code>sys.modules["rutracker"]</code> as a SIDE EFFECT; <code>plugins/</code> is not on <code>sys.path</code> , so bare imports resolved only by luck of the pytest-randomly seed. 12-seed swept after fix
<code>search_stream</code> false-positive	a test loaded <code>routes.py</code> under alias <code>api_routes</code> , re-running the rate-limit decorator and registering the SAME function twice; the allowlist keyed on module path missed the duplicate. Isolation saw 3 endpoints, sweep saw 4 — the

Defect	Root cause
Tracker roster diverged 3 ways	4th a duplicate, not a new endpoint nrm-club.me in the bridge only → merge service fell through to the UNAUTHENTICATED path, saving an HTML 401 body as a .torrent
4 CORS defects	unset vs "" took different code paths (unset DENIED :4200 and all 127.0.0.1); MERGE_SERVICE_PORT hardcoded; wildcard matched whole-string so an operator's * did nothing; whitespace input failed CLOSED to an empty allowlist
11 bare except: across 8 live plugin files	caught KeyboardInterrupt/ SystemExit; diagnostics now on stderr — fd 1 is the nova3 result stream
Cyrillic decode untested	existing tests fed cp1251 bytes but asserted only ASCII titles; nrmclub decodes errors="ignore". 25 tests now assert exact equality against a real library title

Instrument integrity (detectors that could not do their job)

Defect	Root cause
Verdict extractor substring-matched	^ bound only the first alternation group; tail -n1 then took a prose line. Measured: it flipped a PASS into a SKIP. Blast radius corrected: 9 wired gates, 0 currently misreported, 19/267 latent — not the 267 first claimed
Propagation gate bare-literal	grep -qF '11.4.238' satisfied by any mention; mirrors had 0 block-starts

Defect	Root cause
Credentials clause prefix collision	§11.4.109 CONTAINS §11.4.10, so the check passed even with §11.4.10 deleted
My own D1 check carrier-blind	grepped the raw file, so a truthful comment about the historical defect read as live code — it had already forced another agent to reword accurate documentation

Gates validated rather than assumed

check_cm_no_production_mutation_residue (16 arms) and check_cm_healthcheck_covers_served_ports (15 arms) both proved SOUND under mutation, gates left byte-identical. Notable: a mutation stripping the residue scanner's string mask initially SURVIVED all five carrier arms — none had a comment introducer inside a string literal. Five arms no mutation could break were decoration; a sixth with the right shape now catches it (§11.4.194(6)(d)).

Honest boundaries carried forward

- **Reboot survival** remains UNPROVEN end-to-end. Units installed, enabled, valid, linger on — necessary, not sufficient. Host power operations are hard-banned (CONST-033); operator-attended verification required.
- **Firefox / WebKit**: Playwright's bundled browsers were never installed (~/.cache/ms-playwright empty); the Chromium proof used the host snap via executablePath. Driving host Firefox HUNG for 120s, consistent with Playwright requiring its own patched build. Measured, not assumed.
- **Roster presence** is asserted by consumer-agreement, which cannot prove a domain is PRESENT (the test iterates the roster it validates). Being closed.
- **tests/security/ tally is not deterministic** — 9 failed/14 failed/0 failed across three runs, flipping with live-stack availability. A §11.4.50 issue, proven NOT caused by this batch (stash-and-rerun gave an identical tally).